

**BALANGKAS: A GAMIFIED E-LEARNING AND ASSESSMENT SYSTEM FOR
DATA STRUCTURES AND ALGORITHMS**

**A Capstone Proposal
Presented to the Faculty of
College of Information and Communications Technology
STI West Negros University**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

**Vicente E. Pioneta
Lee Ruben H. Valero
Benedict P. Piñon
Dave Venge S. Malunes**

March 2026

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

**TITLE OF RESEARCH: Balangkas: A Gamified E-Learning and Assessment System
For Data Structures and Algorithms**

**NAME OF PROPONENTS: Vicente E. Pioneta Jr.
Lee Ruben H. Valero
Benedict P. Piñon
Dave Venge S. Malunes**

**In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.**

ENDORSED BY:

**RHYAN C. DE LOYOLA
Capstone Adviser**

APPROVED FOR PROPOSAL DEFENSE:

**DANICA S. DUAZO
Capstone Coordinator**

**NOTED BY:
MAE B. LODANA, PhD TM
CICT Dean**

May 2026

APPROVAL SHEET

This thesis titled **Balangkas: A Gamified E-Learning and Assessment System For Data Structures And Algorithms** , prepared and submitted by **Vicente E. Pioneta Jr.; Lee Ruben H. Valero; Benedict P. Piñon ; and Dave Venge S. Malunes**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

RHYAN C. DE LOYOLA

Capstone Adviser

Accepted and approved by the Capstone Review Panel
in partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

MYCA PACIBO

Panel Member

KURT N. BONCALON

Panel Member

ORVILLA V. BALANGUE, MSCS

Lead Panelist

APPROVED:

DANICA S. DUAZO

Capstone Coordinator

MAE B. LODANA, PhD TM

CICT Dean

May 2026

ACKNOWLEDGEMENTS

We would like to express their sincere gratitude to all individuals who contributed to the completion of this study.

First and foremost, we extend our deepest appreciation to our capstone adviser Mr. Rhyann De Loyola for the continuous guidance, support, and valuable insights provided throughout the development of this study. His expertise and encouragement greatly contributed not only in the development of the study but also in our well-being.

We would also like to thank our validators Ms. Cherry Ann Angus, Mr. Jhunrhick Tarre, and Mr. Raymart Abella for their time, effort, and constructive feedback in evaluating the survey for data collection. Their suggestions and professional insights helped improve the quality and reliability of the study.

Special thanks to our capstone review panel, our head panelist, Engr. Orvilla Balangue, our panel members, Ms. Kurt Boncalon and Ms. Myca Pacibo for their valuable recommendations, comments, and guidance, which helped refine and strengthen the study.

Also to our parents and guardians we are always grateful for their unwavering financial support, understanding, and encouragement throughout the development of the study.

To our friends for their moral support, motivation, and inspiration that helped us persevere in developing this study.

Above all, we offer our deepest gratitude to God Almighty for the strength, wisdom, and guidance in overcoming challenges and making this study possible.

ABSTRACT

Title of research: <Research Title>

Researchers: < Researcher's Given Name MI. Family Name>
 < Researcher's Given Name MI. Family Name>
 < Researcher's Given Name MI. Family Name>
 < Researcher's Given Name MI. Family Name>

Degree: Bachelor of Science in <Program>

Date of Completion: <Month year of graduation>

Keywords: <keywords of your research>

The abstract is an important component of your thesis. Presented at the beginning of the thesis, it is likely the first substantive description of your work read by an external examiner. You should view it as an opportunity to set accurate expectations. The abstract is a summary of the whole thesis. It presents all the major elements of your work in a highly condensed form. An abstract is not merely an introduction in the sense of a preface, preamble, or advance organizer that prepares the reader for the thesis. In addition to that function, it must be capable of substituting for the whole thesis when there is insufficient time and space for the full text. Currently, the maximum sizes for abstracts submitted are 150 words to 350 words. Usually a one-pager abstract is the most ideal. To preserve visual coherence, you may wish to limit the abstract for your research to one 1.5-spaced page, about 280 words. The structure of the abstract should mirror the structure of the whole thesis and should represent all its major elements. There should be one or more sentences assigned to summarize each chapter.

In the succeeding paragraphs, there should be no indentations, paragraphs are justified with left alignment. Delete this highlighted section and replace it with your Abstract.

TABLE OF CONTENTS

	Page
Title Page	i
Endorsement Form for Oral Defense	ii
Approval Sheet	iii
Acknowledgments	iv
Abstract	v
Table of Contents	vi
List of Tables	vii
List of Figures	viii
List of Appendices	ix
CHAPTER I – INTRODUCTION	
Background of the Problem	1
Overview of the Current State of the Technology	3
Objectives of the Study	6
Scope and Limitations of the Study	6
Significance of the Study	7
Definition of Terms	8
CHAPTER II - REVIEW OF RELATED LITERATURE AND STUDIES	
Related Literature	10
Related Studies and/or System	13
Synthesis/Research Gap	16
CHAPTER III – METHODOLOGY	
Research Design	19
System Development Methodology	20
Data Gathering	22
Respondents/System Users	23
Hardware and Software Requirements	25
Operational Framework	26
APPENDICES	

LIST OF TABLES

Table		Page
1	<Title of the table>	
2	<Title of the table>	
3	<Title of the table>	
4	<Title of the table>	

LIST OF FIGURES

Figure		Page
1	<Title of the figure>	
2	<Title of the figure>	
3	<Title of the figure>	
4	<Title of the figure>	

LIST OF APPENDICES

Appendix	Page
A Data Sources	33
B Report Layouts/Forms	35
C Written Communications	36
D Interview Guide/Survey Questionnaire	37
E Benefactor's Location Map	43
F Pictures (Data Gathering)	44

CHAPTER I

INTRODUCTION

Background of the Problem

Data Structures and Algorithms (DSA) is a core subjects in Information and Computing programs as it helps students develop the problem-solving skills, logic thinking and formulation, and efficient program design. However, many students find this course difficult because most concepts are abstract and the processes are not easily seen while the program is in runtime. Similar challenges are also experienced by students learning DSA at STI West Negros University. Understanding important concepts becomes harder when students depend only on learning materials and classroom discussions. Many students need more guidance in analyzing and simplifying complex topics, which affects their ability to fully understand the lessons. As a result, their attention may become divided, making the learning process more difficult.

Based on the gathered responses from CICT students, about 90% of the respondents rated Data Structures and Algorithms as moderately difficult to very difficult. This indicates that DSA remains a challenging subject for many students. In addition, around 45% of the respondents shared that learning one DSA concept may take a full day or multiple days before they can apply it independently. The common difficulties experienced by students include remembering algorithmic steps, understanding concepts, applying logic, and visualizing data structures. Specifically, about 49% of the respondents had difficulty remembering processes, followed by 43% who struggled with understanding concepts, 41% with implementing logic, and 39% with visualizing data structures. These findings suggest that students do not only have difficulty memorizing definitions, but also in following step-by-step algorithmic procedures, converting theories into actual code, and imagining how data moves within structures such as recursion, trees, graphs, and other algorithms.

The findings also revealed that students commonly rely on external resources such as online videos, AI tools, websites, and other supplementary materials to better understand DSA. This suggests that existing classroom instruction and available learning materials

may not be sufficient for all learners, especially when dealing with abstract and process-based topics. However, these external resources are often scattered and may not provide structured guidance, immediate feedback, or progress monitoring. At the same time, the respondents showed a generally positive perception toward gamified learning. Approximately 91% of the respondents answered either “Yes” or “Maybe” when asked whether gamification would motivate them to learn DSA. This indicates that most students are open to the use of game-based learning elements as a way to make DSA more engaging, interactive, and easier to approach. The most preferred gamification features were mini games, progress tracking, points system, reward system, badges or achievements, and leaderboards. These findings show the need for a more structured, interactive, and engaging learning platform that can support visualization, practice, feedback, and assessment.

Moreover, while the institution provides a Learning Management System (LMS), its primary use is limited to distributing course materials and submitting assignments. Based on observation, students tend to access the LMS only, when necessary, such as for completing tasks or retrieving files. This indicates that the current system lacks interactive and engaging features that encourage continuous learning and active participation. Studies support that LMS platforms that do not incorporate interactive elements and feedback mechanisms may fail to sustain student engagement and motivation (Kaya & Ercag, 2023).

These challenges highlight a significant need for a more effective learning support system that goes beyond traditional teaching methods and static LMS platforms. There is a need for a system that can provide structured guidance, interactive learning experiences, and opportunities for students to actively engage with DSA concepts. In particular, integrating features such as pre-assessment and post-assessment can help measure student learning progress, while simulation-based activities can support visualization and deeper understanding of algorithmic processes. Features such as immediate feedback, progress tracking, and gamified activities may also help students practice more consistently and monitor their improvement throughout the learning process.

In response to these needs, this study proposes the development of “Balangkas: A Gamified E-Learning and Assessment System for Data Structures and Algorithms” for CICT students at STI West Negros University. The proposed system aims to address the

identified problems by providing an interactive learning environment that integrates instructor-managed content, pre-test and post-test assessments, simulation-based activities, gamification features, and performance tracking. Through the use of gamification and structured learning design, Balangkas is expected to improve student engagement, enhance conceptual understanding, and support learners in mastering DSA concepts more effectively. The findings of this study may contribute to the improvement of teaching strategies and the development of more effective digital learning tools for complex computing subjects.

Overview of the Current State of the Technology

The teaching and learning of Data Structures and Algorithms (DSA) have evolved significantly with the advancement of educational technologies. Traditionally, DSA instruction has been delivered through classroom lectures and programming exercises, where instructors explain concepts while students apply them through coding tasks. While this approach provides foundational knowledge, studies have shown that it is often insufficient for helping students fully understand abstract algorithmic processes, as learners are required to mentally visualize operations without interactive support (Del Mundo & Caballes, 2022). As a result, students frequently experience difficulty in grasping complex topics such as recursion, sorting algorithms, and data structure manipulation.

To address these limitations, Learning Management Systems (LMS) were introduced as platforms for organizing course materials, distributing resources, and managing student assessments. Systems such as Moodle and similar platforms have been widely adopted in higher education institutions to support online and blended learning environments. However, research indicates that LMS platforms are often used primarily for content delivery and administrative purposes rather than as interactive learning tools. In many cases, students engage with the system only when required to submit assignments or access files, limiting its effectiveness in promoting continuous learning and engagement (Kaya & Ercag, 2023).

In recent years, more advanced educational technologies have emerged, including visualization tools and gamified learning systems designed to enhance student interaction

and understanding. Visualization platforms allow learners to observe how algorithms operate step-by-step, making abstract concepts more concrete and easier to comprehend. Meanwhile, gamified systems incorporate elements such as points, levels, challenges, and feedback mechanisms to increase motivation and engagement. Studies have shown that these approaches can significantly improve student participation and learning outcomes when properly integrated into the instructional process (Dichev & Dicheva, 2017; Koivisto & Hamari, 2020).

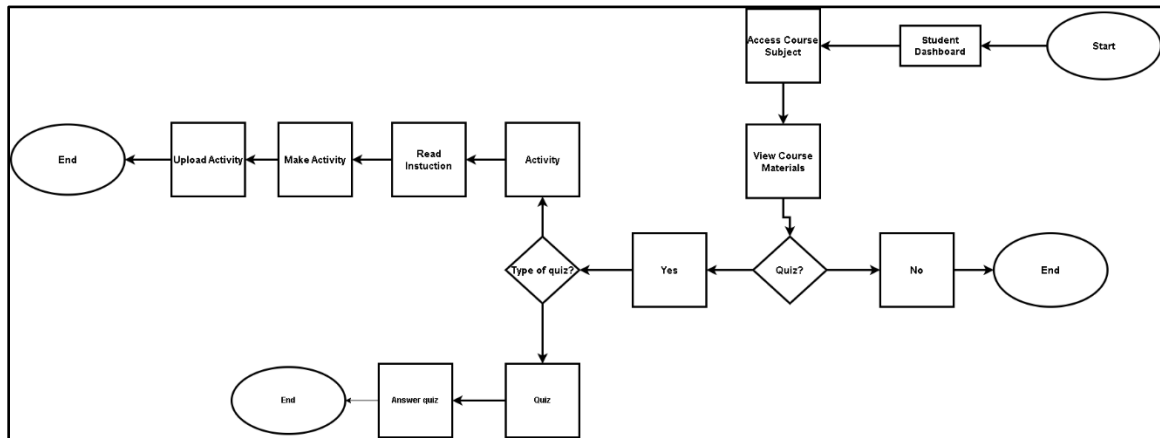
Despite these advancements, existing systems still present several limitations. Many visualization tools focus solely on demonstrating algorithm behavior without integrating structured assessments or instructor-guided learning. On the other hand, gamified platforms often emphasize engagement but lack alignment with course content and formal assessment strategies. Additionally, most systems do not provide a unified environment where instructors can manage lesson content, monitor student progress, and integrate both assessment and interactive learning activities. This fragmentation limits the effectiveness of current technologies in fully addressing the challenges associated with learning DSA.

Furthermore, while these technologies have been available for several years, their implementation in actual classroom settings remains inconsistent. Some institutions continue to rely on traditional teaching methods, while others use LMS platforms that do not fully utilize interactive features. As a result, the core problem of helping students effectively understand and apply DSA concepts persists despite the availability of various technological solutions.

Given the current state of technology, there is a clear need for a more integrated and structured learning system that combines the strengths of existing approaches while addressing their limitations. This study aims to develop “Balangkas: A Gamified E-Learning and Assessment System for Data Structures and Algorithms,” which integrates instructor-managed content, structured pre-test and post-test assessments, and simulation-based learning activities within a single platform. By addressing the gaps in existing technologies, the proposed system seeks to provide a more effective solution for improving student engagement, understanding, and performance in DSA.

Figure 1

Existing Workflow



The proponents have identified problems encountered by the beneficiary in the existing workflow. These are the following:

1. Students experience difficulty in understanding abstract concepts in Data Structures and Algorithms, especially when lessons are presented in a purely theoretical manner.
2. Traditional teaching methods rely heavily on lectures and static materials, which often result in low student engagement.
3. Many students struggle to visualize how algorithms work step-by-step, leading to confusion in implementation.
4. There is a lack of interactive learning tools that allow students to practice Data Structures and Algorithms concepts in a hands-on manner.
5. Instructors have limited platforms where they can easily upload and manage structured learning materials.
6. Students often lack immediate feedback when answering exercises, slowing down their learning progress.
7. Existing systems do not effectively measure students' learning progress before and after instruction.
8. Monitoring student performance across different activities is difficult using manual or traditional methods.

Objectives of the Study

The main objective of the study is to develop BALANGKAS, a gamified e-learning and assessment system that makes Data Structures and Algorithms easier to understand while improving student engagement and academic performance among CICT students.

Specifically, the study aims to:

- To provide interactive learning materials that help students understand key DSA concepts such as arrays, linked lists, stacks, queues, trees, graphs, recursion, sorting, and searching algorithms;
- To integrate gamification features such as points, progress tracking, leaderboards, and challenges to increase students' motivation and engagement in learning DSA;
- To develop visualization features that show how data structures and algorithms work step by step;
- To provide a supplementary learning tool that supports classroom instruction in Data Structures and Algorithms through gamified lessons, interactive activities, visualizations, and assessments; and
- To evaluate the system in terms of usability, functionality, reliability, efficiency, and effectiveness in supporting the learning and assessment of Data Structures and Algorithms.

Scope and Limitations of the Study

This research emphasizes the design and development of an application called BALANGKAS: A Gamified E-Learning and Evaluation System for Data Structures and Algorithms. The system will act as an additional resource to enhance classroom teaching and assist students in better grasping Data Structures and Algorithms via gamified exercises, engaging lessons, algorithm visualization, and evaluation features. It aims to help students grasp key DSA subjects like arrays, linked lists, stacks, queues, trees, graphs, recursion, sorting algorithms, and searching algorithms.

The scope of the system includes the development of learning modules where students can access DSA lessons in a more organized and interactive manner. The system will

also include gamification features such as points, challenges, progress tracking, and leaderboards to encourage student motivation and engagement. The system will also provide visualization features that demonstrate how data structures and algorithms work step by step. Assessment tools such as pre-test, post-test, and DSA concept simulations will also be included to help enhance the level of their understanding. The instructor side of the system will allow teachers to monitor student progress, view assessment results, and evaluate student performance based on completed learning activities.

However, the study has certain limitations. BALANGKAS will only focus on selected topics under Data Structures and Algorithms and will not cover all advanced areas of computer science or topics that cannot be gamified. The system is designed only as a supplementary tool and will not replace classroom discussions, teacher guidance, laboratory activities, or formal instruction. The effectiveness of the system will depend on the students' participation, internet access, device availability, and willingness to use the platform. The system's coding assessment feature may also be limited to predefined activities and test cases, and may not fully evaluate all possible programming solutions. Furthermore, the study will be evaluated only by selected students and instructors from the target institution, so the results may not fully represent the experience of all DSA learners from other schools or programs.

Significance of the Study

This study bring significance to:

Students. This study will help students better understand abstract DSA concepts through visual, interactive, and game-based learning activities. It may also improve motivation, confidence, and independent study habits.

Instructors. This study will provide instructors with a platform for organizing lessons, uploading learning materials, and monitoring student progress in a more efficient way.

The Institution. This study may support the institution in improving the quality of instruction for core computing subjects and in promoting the use of innovative educational technology.

Future Researchers. This study may serve as a reference for future developers and re-searchers who want to create similar systems for other difficult subjects or for broader learning environments

Definition of Terms

To ensure clarity and better understanding of the key concepts used in this study, the following terms are defined operationally and conceptually as applied within the context of the study.

Gamification. This refers to the use of game design elements in non-game contexts to enhance user engagement, motivation, and participation (Dichev & Dicheva, 2017).

E-Learning. This is a mode of education that utilizes digital platforms and electronic technologies to deliver instructional content and facilitate learning (Caporarello, Magni, & Pennarola, 2017).

Data Structures and Algorithms (DSA). This refers to the methods used to organize, store, and process data efficiently, as well as the step-by-step procedures for solving computational problems (Goodrich, Tamassia, & Goldwasser, 2014).

Interactive Visualization. This refers to graphical representations of data or processes that allow users to manipulate and explore information dynamically to enhance understanding (Ware, 2013).

Simulation-Based Learning. This refers to an instructional approach that uses modeled environments or systems to replicate real-world or conceptual processes for educational purposes (Lateef, 2010).

Learning Management System (LMS). This refers to a software platform used to deliver, manage, and track educational courses, instructional materials, and learner performance (Turnbull, Chugh, & Luck, 2021).

Assessment. This refers to the systematic process of evaluating students' knowledge, skills, and learning progress using various tools such as tests, quizzes, and activities (Black & Wiliam, 2009).

Pre-test and Post-test. These are assessment methods used to measure learners' knowledge before and after instruction in order to determine learning gains (Dimitrov & Rumrill, 2003).

Student Engagement. This refers to the level of students' active participation, involvement, and emotional investment in the learning process (Fredricks, Blumenfeld, & Paris, 2004).

Feedback Mechanism. This refers to a process that provides learners with information about their performance to guide improvement and enhance learning outcomes (Hattie & Timperley, 2007).

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the relevant literature, studies, and systems that support the development of the proposed system. It discusses existing research and concepts related to gamification, e-learning, visualization techniques, simulation-based learning, and learning analytics. These areas are important in understanding how modern educational technologies can improve student engagement, motivation, and learning outcomes, especially in challenging computing subjects such as Data Structures and Algorithms (DSA).

2.1 Related Literature

Challenges in Learning Data Structures and Algorithms

Data Structures and Algorithms (DSA) is considered one of the most challenging subjects in computing because it requires abstract thinking, logical reasoning, and the ability to visualize processes mentally. Mtaho and Mselle (2024) explained that DSA creates a high cognitive load for students, which often leads to poor understanding, low retention, and difficulty in applying concepts. They identified the abstract nature of the topic, weak instructional methods, lack of learning materials, low conceptual understanding, and low motivation as common causes of difficulty. In the same way, Tilanterä (2023) noted that learners often struggle with algorithmic concepts because they require deep mental processing and continuous practice. Hao et al. (2023) also emphasized that programming and algorithm-related tasks can overwhelm learners when the cognitive demands are too high.

In the Philippine setting, similar difficulties were observed in computing education. Studies on gamified and interactive learning showed that students often perform better when lessons are presented in a more engaging and understandable way. This suggests that DSA, whether in foreign or local contexts, remains a difficult subject that needs better teaching approaches to improve student learning.

Gamification in Education

Gamification has become a widely used strategy in education because it helps increase motivation, participation, and learner engagement. Kurt (2023) explained that gamification applies game mechanics such as points, rewards, and competition in non-game settings to create more meaningful learning experiences. Hamari, Koivisto, and Sarsa (2014) also found that game elements such as points, badges, rewards, and leaderboards can positively influence student participation and learning experiences. These studies show that gamification can help maintain learner interest and support progress in difficult subjects like DSA.

Local studies support the same idea. Duterte (2024) found that gamification elements such as points, badges, leaderboards, and collaborative challenges improved academic performance and motivation among undergraduate students in Manila. Domingo, Bautista, and Fajardo (2025) also reported that gamified e-learning platforms improved engagement and academic performance in Philippine higher education institutions. Likewise, Gabatin et al. (2025) and Cajilla and Bug-os (2022) found that gamification had a favorable effect on student satisfaction and learning outcomes. These foreign and local findings both show that gamification is an effective way to make learning more engaging and enjoyable.

Simulation-Based and Interactive Learning

Simulation-based learning and interactive activities also play an important role in helping students understand abstract concepts. Lamrani and Abdelwahed (2020) found that game-based and simulation-based learning improves learner participation and conceptual understanding by encouraging active interaction with educational content. Instead of only reading or listening, learners are able to explore concepts more meaningfully through experience. This approach is especially useful in DSA because many of its processes, such as sorting, searching, and traversal, are difficult to understand without visualization.

This is also supported by local studies. Guisadio et al. (2025) reported that gamified elements such as quests, levels, and rewards positively affected learner motivation, engagement, and performance in Philippine educational settings. These findings show that interactive and simulation-based approaches can help students better understand difficult topics by making lessons more engaging and easier to follow.

Visualization and Interactive Learning in Computer Science Education

Visualization is one of the most effective ways to reduce the difficulty of learning abstract computer science concepts. Theoharis, Papaioannou, Platis, and Patrikalakis (2008) stated that graphical and visual representations improve comprehension by helping learners understand relationships, movements, and transformations in a system. This is especially useful in DSA, where many operations are not directly visible through text-based instruction alone. Osman and Elmusharaf (2014) also found that combining algorithm and program animation in teaching data structure courses improved student performance and understanding. Students exposed to animation performed better than those who learned through traditional methods.

Local research shows the same pattern. Nob, Roble, and Lomibao (2024) reported that gamification improves learner engagement, motivation, academic performance, and higher-order thinking skills. Pelegro (2022) also explained that gamification makes online learning more interactive and enjoyable. Landicho et al. (2017) demonstrated through the Hortari application that gamification can improve student engagement and motivation in higher education. These studies support the idea that visual and interactive learning tools are helpful in improving understanding, especially in technical subjects like DSA.

Learning Analytics in Educational Systems

Learning analytics is another important component in educational systems because it helps instructors monitor student performance and provide timely feedback. Lepouras, Katifori, Vassilakis, Antoniou, and Platis (2014) emphasized that learning analytics can help identify weaknesses, measure progress, and evaluate the effectiveness of learning materials and systems. This supports the inclusion of performance tracking in Balangkas so that instructors can better monitor student progress and learning outcomes.

In the local context, studies on gamified systems also highlight the value of tracking learner performance to improve academic results. The positive findings from Philippine studies on gamification suggest that monitoring student progress can help teachers and developers understand how learners respond to the system and where improvements are needed. This makes learning analytics a useful feature for supporting student success.

2.2 Related Studies and/or Systems

Review Related Systems/Studies

This section presents a review of existing systems and studies relevant to the development of the proposed system. The review covers works on gamification in education, the challenges associated with learning Data Structures and Algorithms (DSA), visualization and simulation-based learning, as well as existing systems both locally and internationally. These prior works were examined to establish the theoretical and empirical foundations of the study, identify gaps in existing approaches, and highlight the need for an integrated and interactive learning platform for DSA learners.

Table 1

Comparison of Related Systems/Applications

System Name & Criteria	Visu- Algo	AGBL	Stack GBL	Active Learning Game (Dicheva & Hodge	Code It!	Hor-tari	CSS Quest	LogIO	Bal- angkas (Pro- posed)
Platform	Web	Web	Web	Web	Web	Web	Web	Web	Web
Target Sub- ject	Data Struc- tures & Algo- rithms	Data Struc- tures	Data Struc- tures (Stack)	Data Struc- tures (Stack)	Itera- tive Pro- gram- ming	Gen- eral	Com- puter Sys- tem Ser- vic- ing	Digi- tal Logic Gates	Data Struc- tures & Algo- rithms
Gamification Elements	No	Partial	Partial	Yes	Yes	Yes	Yes	Yes	Yes
Points, Badges & Leader- boards	No	No	No	Partial	Yes	Yes	Yes	Yes	Yes
Algorithm Visualization	Yes	Yes	Yes	Partial	No	No	No	Partial	Yes
Simulation- Based Activ- ities	Partial	Yes	Yes	Yes	Yes	No	No	No	Yes

Pre-test & Post-test Assessment	No	No	Yes	Yes	No	No	No	No	Yes
Instructor Content Management	No	No	No	No	No	No	No	No	Yes
Performance Tracking & Analytics	Partial	No	No	No	No	No	Partial	Partial	Yes
Immediate Feedback Mechanism	Partial	Partial	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Methodology	Iterative Development	Quasi-Experimental	Pre-Experimental	Quasi-Experimental	Experimental	Developmental	Iterative & Incremental	Developmental-Evaluative	DevOps CI/CD

Web Platform. The reviewed systems show different platform approaches. VisuAlgo, AGBL, Stack GBL, Active Learning Game, Code It!, Hortari, CSS Quest, and Balangkas are web-based, while LogIO is available as both a web and mobile platform. In this study, Balangkas is developed as a web-based system to ensure easier access for CICT students using common browsers and devices without requiring installation.

Target Subject. The target subjects of the reviewed systems also vary. VisuAlgo focuses on Data Structures and Algorithms, AGBL and Stack GBL focus on Data Structures and stacks, while Code It! is centered on iterative programming. Hortari is designed for general use, CSS Quest for Computer System Servicing, and LogIO for Digital Logic Gates. Balangkas is specifically developed for Data Structures and Algorithms, making it directly aligned with the needs of students who struggle with abstract algorithmic concepts.

Gamification Elements. Some of the reviewed systems already include gamification, but not all use it fully. VisuAlgo does not include gamification, while AGBL and Stack GBL only apply it partially. Active Learning Game, Code It!, Hortari, CSS Quest,

LogIO, and Balangkas include gamification elements. In this study, Balangkas fully integrates gamification to support learner motivation, engagement, and participation in a more structured way.

Points, Badges & Leaderboards. VisuAlgo, AGBL, and Stack GBL do not use points, badges, and leaderboards. Active Learning Game includes them only partially, while Code It!, Hortari, CSS Quest, LogIO, and Balangkas include these features. Balangkas uses these elements to encourage active participation, reward progress, and sustain learner interest throughout the learning process.

Algorithm Visualization. VisuAlgo, AGBL, and Stack GBL provide algorithm visualization, while Active Learning Game provides it only partially. Code It!, Hortari, and CSS Quest do not include algorithm visualization, and LogIO offers only partial support. Balangkas includes full algorithm visualization to help students understand step-by-step processes in Data Structures and Algorithms more clearly.

Simulation-Based Activities. VisuAlgo provides only partial support for simulation-based activities, while AGBL, Stack GBL, Active Learning Game, and Code It! include them. Hortari, CSS Quest, and LogIO do not provide simulation-based activities. Balangkas includes simulation-based learning activities to give students hands-on practice and a more interactive learning experience.

Pre-test & Post-test Assessment. VisuAlgo and AGBL do not include pre-test and post-test assessments. Stack GBL and Active Learning Game include them, while Code It!, Hortari, CSS Quest, and LogIO do not. Balangkas includes pre-test and post-test assessments to measure students' learning progress before and after instruction.

Instructor Content Management. None of the reviewed systems except Balangkas provide instructor content management. This feature is important because it allows instructors to upload, organize, and manage learning materials within one platform. Balangkas includes this functionality to support structured lesson delivery and better classroom management.

Performance Tracking & Analytics. VisuAlgo, CSS Quest, and LogIO only partially support performance tracking and analytics, while AGBL, Stack GBL, Active Learn-

ing Game, Code It!, and Hortari do not provide this feature. Balangkas includes full performance tracking and analytics so that instructors can monitor student progress and identify areas that need improvement.

Immediate Feedback Mechanism. Most of the reviewed systems include immediate feedback mechanisms, although VisuAlgo, AGBL, and Active Learning Game only provide them partially. Stack GBL, Code It!, Hortari, CSS Quest, LogIO, and Balangkas include this feature. In Balangkas, immediate feedback is used to help students quickly understand their mistakes and improve learning during activities and assessments.

2.3 Synthesis/Research Gap

The reviewed literature and studies consistently demonstrate that gamification, interactive visualization, and simulation-based learning are effective strategies for improving student engagement, motivation, and academic performance, particularly in technically demanding subjects such as Data Structures and Algorithms. Internationally, studies confirm that gamified elements such as points, badges, leaderboards, and rewards positively influence learning outcomes (Hamari et al., 2014; Domínguez-Díaz et al., 2013; Abu-Hammad & Hamtini, 2023). In the Philippine context, local research likewise affirms that gamification improves motivation and engagement among Filipino learners at various educational levels (Duterte, 2024; Guisadio et al., 2025; Domingo et al., 2025). Visualization tools have further been shown to reduce cognitive load and improve conceptual understanding of abstract algorithmic processes (Osman & Elmusharaf, 2014; Theoharis et al., 2008; Halim, 2015).

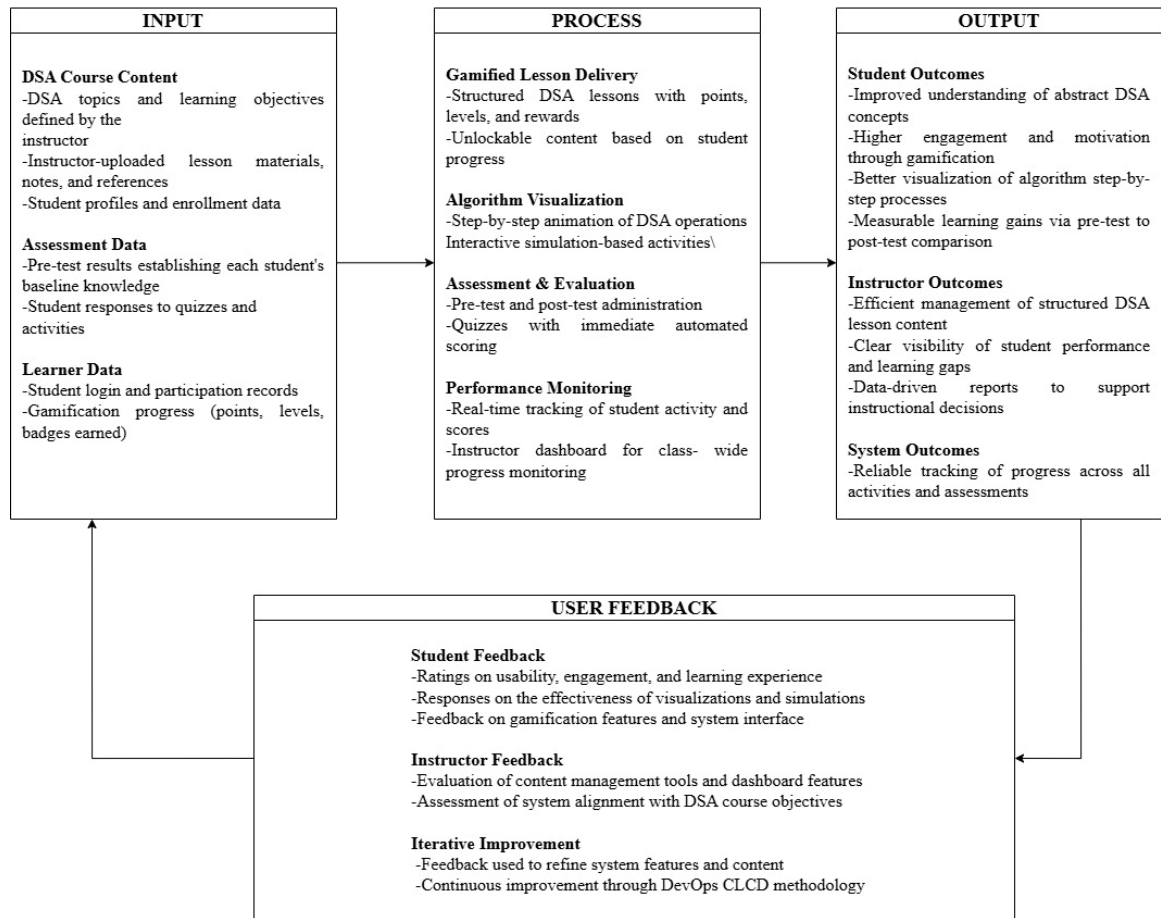
However, the synthesis of existing works reveals a significant research and development gap. Most reviewed systems address only one or two dimensions of effective DSA instruction. Visualization platforms such as VisuAlgo improve conceptual understanding but lack gamification and structured assessment features. Gamified systems such as Code It! and CSS Quest improve engagement but were not developed for DSA and do not integrate step-by-step algorithm visualization. Other systems such as AGBL and the Stack game-based learning tools provide gamified DSA experiences but do not include instruc-

tor-managed content delivery or performance analytics. Furthermore, no existing local system has been developed to address the specific needs of DSA learners in the CICT program at STI West Negros University.

Balankas directly addresses this gap by proposing a unified, gamified e-learning and assessment system that integrates all of the following components within a single platform: instructor-managed lesson delivery, step-by-step algorithm visualization, simulation-based learning activities, pre-test and post-test assessments, gamification elements, and performance tracking. By bridging the identified gap between engagement-focused and content-focused systems, Balankas seeks to provide a more effective, complete, and contextually relevant solution for improving student understanding and performance in Data Structures and Algorithms

2.4 Conceptual Framework

Figure2.



CHAPTER III

METHODOLOGY

This chapter presents the methodology used in the development of the proposed system, BALANGKAS. It discusses the methods, procedures, and techniques employed to address the identified learning difficulties encountered by students in Data Structures and Algorithms (DSA). The chapter also explains how the researchers gathered and analyzed the necessary data to support the development of the system.

Based on the survey conducted, students commonly experience difficulties in understanding DSA concepts, implementing logic, remembering algorithmic processes, and visualizing complex structures such as recursion, trees, and graphs. The findings also revealed that many students rely on external learning resources and expressed positive perceptions toward gamified learning approaches. In response to these challenges, the proposed system aims to provide a more interactive, engaging, and learner-centered platform through the integration of gamification features, visual learning tools, and assessment mechanisms.

Furthermore, this chapter describes the research design, respondents of the study, data gathering procedures, instruments used, system development methodology, software and hardware requirements, database design, system architecture, and testing procedures. These sections are interconnected and collectively demonstrate how the proposed system was carefully planned, developed, and evaluated to ensure that it effectively addresses the objectives of the study and improves students' learning experience in DSA.

3.1 RESEARCH DESIGN

This study will use a developmental mixed-method research design. The developmental research design is appropriate because the study focuses on the design, development, and evaluation of BALANGKAS: A Gamified E-Learning and Assessment System for Data Structures and Algorithms. It involves identifying the learning difficulties of students in DSA, developing a system based on these needs, and evaluating the system's features, usability, and effectiveness as a supplementary learning tool.

The study will also use a mixed-method approach because it will gather both quantitative and qualitative data. Quantitative data will be collected through survey responses, rating scales, and system evaluation results to measure the students' level of difficulty, motivation, and acceptance of the system. Qualitative data will be gathered through open-ended responses, feedback, and comments from students and instructors to better understand their experiences, suggestions, and challenges in using the system. Through this design, the study can provide a more complete understanding of how BALANGKAS can support the teaching and learning of Data Structures and Algorithms.

After the development, the system will also be tested and assessed using the ISO/IEC 25010 Software Quality Model to evaluate its overall quality. The evaluation will focus on selected quality characteristics such as functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability. These criteria will help determine whether BALANGKAS meets the required software standards and effectively supports students and instructors in learning and assessing Data Structures and Algorithms.

3.2 SYSTEM DEVELOPMENT METHODOLOGY

The study adopts the DevOps methodology in developing the proposed system, *Balangkas*. DevOps is a software development approach that integrates development and operations practices into a continuous, iterative cycle — covering planning, coding, building, testing, releasing, deploying, operating, and monitoring. It is well-suited for this project because the gamified learning system requires ongoing refinement of its interactive features, quick response to testing feedback, and consistent coordination among team members throughout the development process.

The DevOps workflow is divided into two sides: the Dev Side, which covers the activities directly involved in building the system, and the Ops Side, which handles everything from releasing the finished build to keeping the live system running and monitored.

Development

Plan. The team used MS Teams for communication, meeting coordination, and adviser consultations throughout the project. GitHub Issues served as the task tracker, where

features, bugs, and development tasks were logged and assigned to team members on a per-sprint basis.

Code. Visual Studio Code (VS Code) was the team's primary development environment, chosen for its lightweight build and strong extension support for JavaScript, React, and Tailwind CSS. Git handled local version control, allowing developers to track changes and manage branches independently. GitHub served as the remote repository where all code was stored, reviewed through pull requests, and merged — supporting parallel development across the team.

Build. Node.js provided the runtime environment for executing JavaScript outside the browser and running local development servers. npm managed project dependencies and executed build scripts, including npm run build, to compile the application into a production-ready bundle. React was chosen for its component-based architecture and efficient state management, both of which are well-suited for the platform's dynamic and interactive pages. PhaserJS was selected as the game framework for rendering DSA simulations and algorithm visualizations directly in the browser, requiring no additional installation from the user. Tailwind CSS handled responsive styling through utility-based classes applied directly within components, eliminating the need for large separate stylesheet files.

Test. Manual testing was conducted using the web browser to verify the functionality, layout, and behavior of all system features. Chrome DevTools was used to inspect UI elements, debug JavaScript errors, monitor network requests, and test responsiveness across different screen sizes.

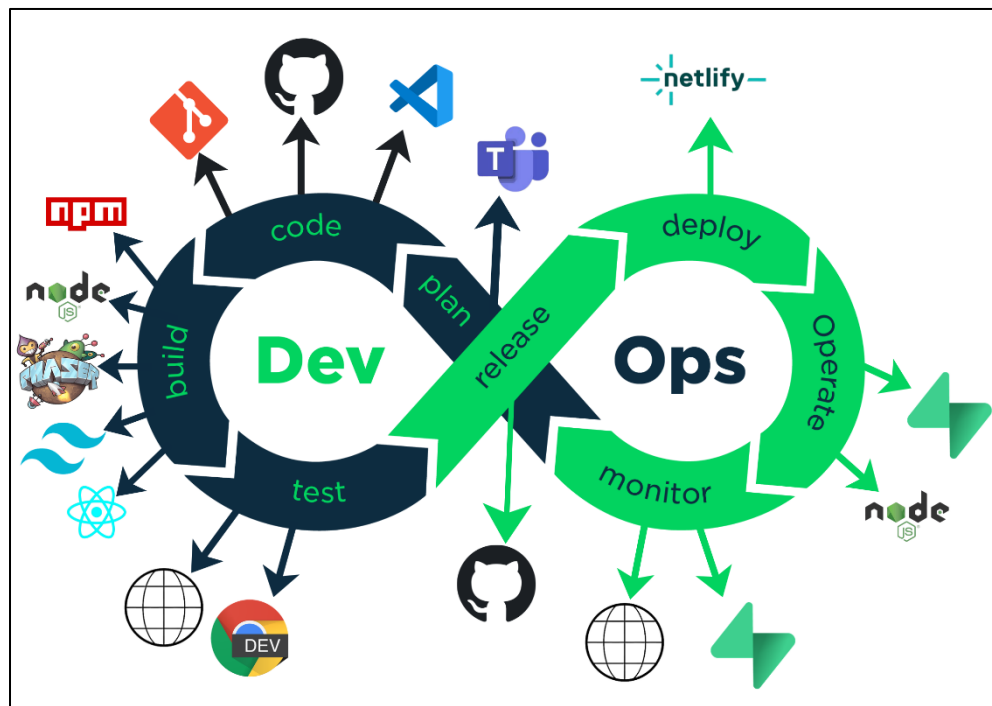
Operations

Release. The release phase is triggered automatically when a developer pushes finalized code to the main branch on GitHub. This push integrates directly with Netlify, initiating an automatic deployment without any manual release step — reducing human error and speeding up delivery.

Deploy. Netlify hosts the production build of Balangkas. Every push to the main branch on GitHub prompts Netlify to pull the latest build and deploy it to a live URL. This continuous deployment setup ensures users always have access to the most recent version of the system without requiring manual server management.

Operate. Supabase serves as the backend-as-a-service layer of the system, providing a PostgreSQL database, user authentication, and a RESTful API that handles all data transactions — including student accounts, progress tracking, scores, and instructor-managed content. Node.js supports server-side scripting and manages API communication between the frontend and Supabase.

Monitor. System monitoring is carried out by reviewing Supabase dashboard logs to track database activity, API calls, and authentication events. The browser console is used to catch and inspect runtime errors in the live production environment, allowing the team to identify and address issues reported during actual use.



3.3 DATA GATHERING

Data gathering is necessary in order to determine whether existing problems are present in the teaching and learning of Data Structures and Algorithms. The responses collected from both instructors and students will serve as the primary basis for identifying the difficulties that the proposed system must address. The researchers will gather data from the instructors through interviews and from the students through survey questionnaires.

These instruments are suitable because they allow the researchers to collect both descriptive and numerical information relevant to the study.

The interview guide for the DSA instructors will be used to obtain qualitative information regarding classroom challenges, student difficulties, and instructional concerns. The survey questionnaire for the students will be used to collect quantitative data on their experiences, understanding, engagement, and perceived difficulties in the subject. After the responses are collected, the researchers will count the responses, tabulate the results, and compute the mean in order to identify the most common issues and tendencies reflected in the data.

The collected information will support the study because it will provide evidence of the actual needs of the students and the observations of the instructors. These findings will guide the researchers in identifying the features and improvements that should be included in the gamified learning system. In this way, the data gathering procedure directly contributes to the development of a solution that is aligned with the real learning problems experienced in DSA.

3.4 RESPONDENTS/SYSTEM USERS

The primary respondents of the study are second-year students of the College of Information and Communications Technology (CICT) at STI West Negros University who have already taken Data Structures and Algorithms. These students will also serve as the main users and testers of the proposed gamified e-learning and assessment system. They are considered appropriate respondents because they have recently encountered the subject and can provide relevant insights regarding the level of difficulty, learning challenges, and overall experience in studying DSA.

Third-year and fourth-year CICT students were also included to widen the range of responses and support data reliability. While second-year students formed the primary group, higher-year students who had previously taken DSA were able to contribute relevant responses drawn from their own experience with the subject.

The estimated total population of second-year Information Technology students is 400. To determine the required sample size, the researchers applied Slovin's formula:

$$n = N / (1 + Ne^2)$$

Where:

- n = sample size
- N = total population (400)
- e = margin of error (0.05 or 5%)

Solving for n :

$$n = 400 / (1 + 400 \times 0.05^2) \quad n = 400 / (1 + 400 \times 0.0025) \quad n = 400 / (1 + 1) \quad n = 400 / 2 \quad n = 200$$

The computed minimum sample size is 200 respondents. The study collected a total of 210 responses, distributed as follows: 163 second-year students, 41 third-year students, and 6 fourth-year students. This exceeded the minimum required sample, which strengthens the reliability of the gathered data.

Among the 163 second-year respondents, thematic saturation was observed at approximately 20 to 30 responses, where recurring patterns and similar perceptions regarding DSA difficulties, learning obstacles, and gamified learning preferences became repetitive and no substantially new themes emerged. The continued collection beyond that point served to confirm the consistency of findings rather than uncover new ones.

The selected respondents are suitable for the study because they are CICT students of STI West Negros University who have already studied Data Structures and Algorithms and can provide meaningful feedback on the effectiveness of the proposed system in improving understanding, engagement, and learning experience.

3.5 HARDWARE AND SOFTWARE REQUIREMENTS

VSCODE - We used this as our main IDE as this provides and supports extension for development.

Github - We used this for version control and group collaboration

Browser - Since we are developing a web based game browser is a necessary tool for testing.

DEVELOPMENT TOOLS

React Native - We chose to use React for the dynamic characteristic of React when it comes to UI updates rendering things like SPA(Single Page Application) behaviour through client-side routing and state management.

PhaserJS - Since our platform is web, we decided to use the framework PhaserJS. It's lightweight and is browser-based it also doesn't require any installations like every other game engines making its setup easier.

Tailwind - We use Tailwind on some pages that might need responsiveness across different screen resolutions. Its utility-based styling also speeds up development by allowing styles to be applied directly within components without creating large separate CSS files.

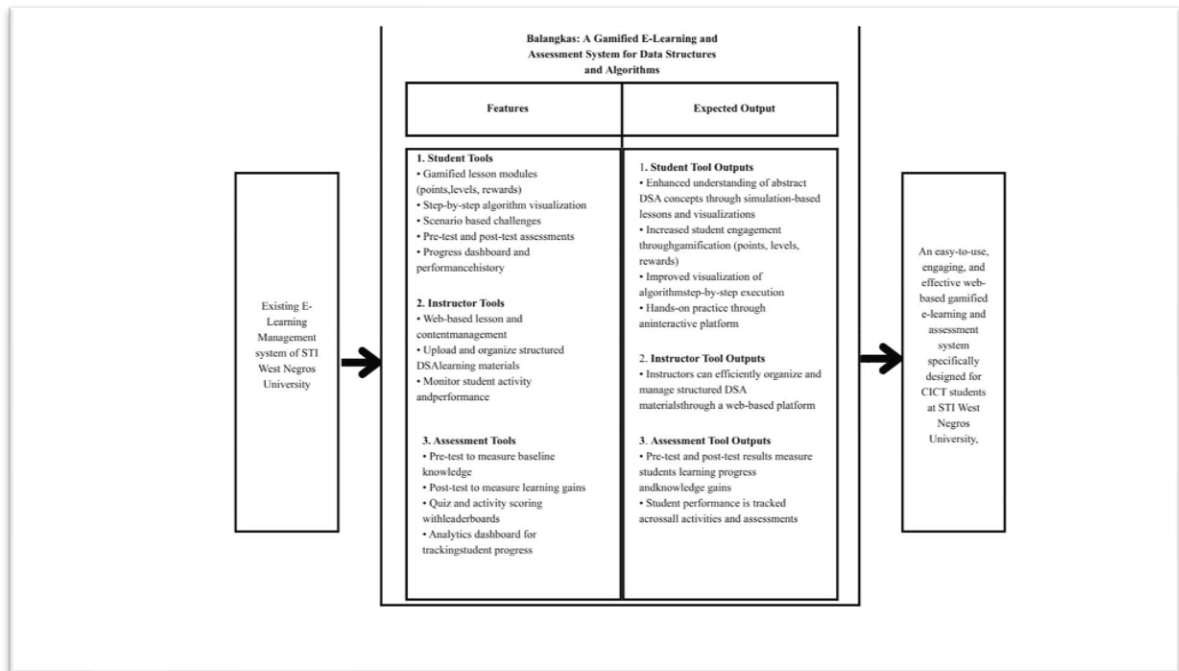
PROGRAMMING LANGUAGES

We used JavaScript as the main programming language for both the frontend interface and game logic of the platform as it is usually used to handle interactive features, user input, content rendering, and communication between different components of the system. JavaScript also works well with React and Phaser which is very suitable to web-based application and games

HARDWARE SPECIFICATIONS

The minimum hardware requirements for the system include a dual-core processor running at 1.8 GHz, 4 GB of RAM, and at least 200 MB of free storage space for browser cache and assets. The system also requires integrated graphics with WebGL 1 support, a display resolution of 1366×768, and a stable broadband internet connection with a minimum speed of 5 Mbps. For better performance and a smoother user experience, the recommended hardware specifications include a quad-core processor running at 2.5 GHz or higher, 8 GB of RAM or more, and at least 1 GB of free storage space. In addition, a modern integrated or dedicated GPU with WebGL 2 support, a display resolution of 1920×1080 or higher, and a high-speed broadband connection of 10 Mbps or higher are recommended.

3.6 OPERATIONAL FRAMEWORK



REFERENCES

- Abu-Hammad, A., & Hamtini, T. M. (2023). Gamification in online programming education: Effects on student engagement and motivation. *International Journal of Emerging Technologies in Learning*, 18(4), 45–60.
- Black, P., & Wiliam, D. (2009). Developing the theory of formative assessment. *Educational Assessment, Evaluation and Accountability*, 21(1), 5–31.
- Cajilla, R. J., & Bug-os, J. A. (2022). Gamified learning management system and students' academic achievement in Physical Science. *International Journal of Multidisciplinary Research and Analysis*, 5(9).
- Caporarello, L., Magni, M., & Pennarola, F. (2017). Learning and e-learning in organizations. *International Journal of Advanced Corporate Learning*, 10(3), 4–10.
- Del Mundo, R. C., & Caballes, D. G. (2022). Visualization techniques in teaching Data Structures and Algorithms among computing students. *International Journal of Computing Sciences Research*, 6, 1300–1312.
- Dichev, C., & Dicheva, D. (2016). Gamifying education: What is known, what is believed and what remains uncertain. *International Journal of Educational Technology in Higher Education*, 13(1), 1–36.
- Dichev, C., & Dicheva, D. (2017). Gamification in education: Where are we in 2017? In *Proceedings of the 9th International Conference on Education and New Learning Technologies* (pp. 1445–1454).
- Dimitrov, D. M., & Rumrill, P. D. (2003). Pretest-posttest designs and measurement of change. *Work*, 20(2), 159–165.

Domingo, R., Bautista, M., & Fajardo, J. (2025). Impact of gamified e-learning platforms on student engagement and academic performance in higher education institutions in the Philippines. *Philippine Journal of Educational Technology*, 7(1), 55–72.

Domínguez-Díaz, A., Saenz-de-Navarrete, J., De-Marcos, L., Fernández-Sanz, L., Pagés, C., & Martínez-Herráiz, J. (2013). Gamifying learning experiences: Practical implications and outcomes. *Computers & Education*, 63, 380–392.

Duterte, M. L. (2024). Gamification in online learning environments among undergraduate students in Manila. *Asia Pacific Journal of Multidisciplinary Research*, 12(2), 34–48.

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. *Review of Educational Research*, 74(1), 59–109.

Gabatin, J. P., Reyes, M. T., & Ramos, A. D. (2025). Effectiveness of gamification on student satisfaction and learning outcomes at Mapua Malayan Colleges Laguna. *Mapúa Journal of Digital Learning*, 3(1), 21–39.

Goodrich, M. T., Tamassia, R., & Goldwasser, M. H. (2014). *Data structures and algorithms in Java* (6th ed.). Wiley.

Guisadio, R. A., Santos, P. L., & Villanueva, K. M. (2025). Gamified elements in Philippine education: A systematic review. *Philippine Social Science Journal*, 8(1), 88–104.

Halim, S. (2015). VisuAlgo: Visualizing data structures and algorithms through animation. *ACM Inroads*, 6(4), 50–55.

Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does gamification work? A literature review of empirical studies on gamification. In *Proceedings of the 47th Hawaii International Conference on System Sciences* (pp. 3025–3034).

- Hao, Q., Smith, D. H., Ding, L., & Ko, A. J. (2023). Cognitive challenges in algorithm learning and programming education. *ACM Transactions on Computing Education*, 23(2), 1–25.
- Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, 77(1), 81–112.
- Kaya, T., & Ercag, E. (2023). The effects of learning management systems on student engagement in higher education. *Education and Information Technologies*, 28(5), 5111–5130.
- Khlaisang, J., & Mingsiritham, K. (2016). Engaging learners in ASEAN through technology-supported learning environments. *International Journal of Information and Education Technology*, 6(5), 350–355.
- Koivisto, J., & Hamari, J. (2020). The rise of motivational information systems: A review of gamification research. *International Journal of Information Management*, 45, 191–210.
- Kurt, S. (2023). Gamification in education: Benefits and challenges. *Educational Technology Insights Journal*, 9(2), 14–22.
- Lamrani, R., & Abdelwahed, E. H. (2020). Game-based learning and simulation in education: A literature review. *Education and Information Technologies*, 25(6), 5311–5337.
- Landicho, A. B., Cabanlit, J. R., & Mendezabal, M. J. (2017). Hortari: A gamified application for student engagement in higher education. *Asia Pacific Journal of Education, Arts and Sciences*, 4(3), 23–30.
- Lateef, F. (2010). Simulation-based learning: Just like the real thing. *Journal of Emergencies, Trauma and Shock*, 3(4), 348–352.

Lepouras, G., Katifori, A., Vassilakis, C., Antoniou, A., & Platis, M. (2014). Learning analytics and educational data mining in practice. *International Journal of Technology Enhanced Learning*, 6(2), 111–128.

Mtaho, A. B., & Mselle, L. J. (2024). Challenges in learning data structures and algorithms: A cognitive load theory perspective. *International Journal of Computer Science Education in Schools*, 8(1), 45–61.

Musina, A. (2024). Personalized learning approaches in teaching algorithms and data structures. *International Journal of Emerging Technologies in Learning*, 19(3), 75–89.

Nob, J. R., Roble, A. C., & Lomibao, H. J. (2024). Gamification and learner engagement in Philippine education: A literature review. *Philippine Journal of Educational Research*, 14(1), 66–82.

Osman, A., & Elmusharaf, K. (2014). Effectiveness of algorithm and program animation in teaching data structures courses. *International Journal of Computer Applications*, 89(7), 12–18.

Pelegro, R. S. (2022). Gamification and student motivation in online educational environments in the Philippines. *Philippine E-Journal of Applied Research and Development*, 9(2), 45–57.

Theoharis, T., Papaioannou, G., Platis, N., & Patrikalakis, N. (2008). *Graphics and visualization: Principles and algorithms*. A K Peters.

Tilanterä, M. (2023). Student misconceptions and cognitive challenges in learning algorithms. *Computer Science Education*, 33(1), 1–20.

Turnbull, D., Chugh, R., & Luck, J. (2021). Learning management systems: A review of the research methodology literature in higher education. *International Journal of Educational Technology in Higher Education*, 18(1), 1–24.

Ware, C. (2013). *Information visualization: Perception for design* (3rd ed.). Morgan Kaufmann.

Xie, H., & Pamintuan, C. F. (2025). Digital learning engagement among ASEAN students in higher education. *Asian Journal of Distance Education*, 20(1), 90–108.

APPENDICES

APPENDIX A

DATA SOURCES

RESP_ID	YEAR_LEVEL	STATUS	PROGRAM	ESS	N	Y	STER	B1_ARRAYS	T	B1_STACKS	B1_QUEUES	B1_TREES	B1_HEAPs	B1_SETs	B1_MAPs	B1
1	2	Regular	BSIT	3	2	3	2	2	3	1	1	2	1	2	3	
2	2	Regular	BSIT	3	1	2	4	2	2	2	2	2	2	2	2	
3	2	Regular	BSIT	3	2	3	2	3	3	4	2	1	2	2	1	
4	2	Regular	BSIT	3	1	3	4	2	2	2	1	1	3	2	1	
5	2	Regular	BSIT	2	1	3	2	3	3	3	3	3	3	3	3	
6	2	Regular	BSIT	3	1	2	3	2	2	2	2	2	2	2	2	
7	2	Regular	BSIT	3	2	3	2	2	2	2	2	2	2	2	2	
8	2	Regular	BSIT	3	2	3	4	1	2	2	2	3	3	2	4	
9	2	Regular	BSIT	2	2	3	2	2	2	2	2	3	3	1	4	
10	2	Regular	BSIT	2	1	5	3	4	4	4	4	3	4	4	4	
11	2	Regular	BSIT	3	1	4	4	4	5	5	5	5	4	2	4	
12	2	Regular	BSIT	3	1	4	4	3	4	5	4	4	3	2	3	
13	2	Regular	BSIT	3	2	3	2	3	3	3	3	3	3	3	3	
14	2	Regular	BSIT	2	1	3	2	3	3	2	2	3	3	3	3	
15	2	Regular	BSIT	2	1	4	4	3	2	2	2	2	2	2	3	
16	2	Regular	BSIT	2	1	3	4	4	3	4	4	4	3	3	4	
17	2	Irregular	BSIT		1			2	2	2	2	2	1	1	2	
18	2	Regular	BSIT	3	1	3	1	2	2	3	3	2	4	2	5	
19	2	Regular	BSIT	3	2	3	4	3	3	3	3	3	3	3	3	

B1_GRAPHs	N	N	MS	ANDING	B2_APPLYING	NG	ERING	NTING	B2_OTHERS	OUTSIDE	L_RESOURCE	S	L_MOVEMEN	MPLEMENT	ER_STEPS	ROBL
START ENCODING DATA FROM ROW 3 BELOW																
3	4	3	3	1	1	1	0	0	0	5	4	2	2	2	3	2
2	2	2	0	0	0	1	0	0	0	4	4	3	3	2	3	3
2	3	2	1	1	0	0	0	0	0	3	5	3	4	3	4	4
3	5	4	3	0	0	1	0	0	0	3	4	2	3	3	2	4
3	3	3	3	0	0	1	0	0	0	5	5	1	1	1	1	1
2	2	3	2	1	1	1	1	1	1	3	3	3	2	3	3	3
2	2	2	2	0	1	0	1	1	0	3	5	3	3	3	3	3
3	2	2	3	1	1	0	0	1	0	3	3	4	3	4	4	3
4	2	2	2	1	0	0	0	1	0	2	2	3	3	4	3	4
4	4	4	5	0	0	1	0	0	0	3	4	3	4	3	4	3
4	5	5	3	1	1	1	1	1	0	4	5	5	4	4	4	5
3	4	4	3	1	1	1	0	0	0	4	3	3	4	3	4	3
3	3	3	3	1	0	0	0	0	0	4	3	3	3	4	4	3
2	3	4	3	0	0	0	1	0	0	3	4	4	2	3	3	3
3	3	4	2	1	1	1	1	0	3	2	2	2	3	3	5	5
4	5	5	5	0	1	1	0	1	0	1	4	4	4	5	5	4
3	3	3	3	0	0	0	1	1	0	2	2	3	2	3	4	2
4	5	4	3	1	1	0	1	0	0	2	5	2	3	3	3	4
3	3	3	3	0	1	0	1	1	0	3	4	4	2	4	4	3

PROBLEMS	OCKAGE	NT_STRUCTU	ERNAL	ALGO	UNDATION	M_USED	M_SPECIFY	MOTIVATE	B6_POINTS	B6_BADGES	OARDS	ES	S	B6_REWARDS	B6_OTHERS	_MC
2	4	2	2	3	2	1	N/A	1	1	0	0	1	0	0	0	
3	3	3	3	3	3	1	Youtube	2	0	1	0	1	1	1	0	0
4	5	2	3	4	3	1	Website	3	0	0	1	0	0	0	0	1
4	3	5	5	4	4	1	Youtube, Stack Overflow	1	0	1	1	0	0	1	0	1
1	5	5	5	5	5	0		1	0	1	0	0	0	0	0	1
3	3	3	2	3	3	1	w3schools	3	1	1	1	1	1	1	0	0
3	3	3	4	3	3	0		3	0	1	1	1	1	1	0	0
3	3	3	3	4	3	0		3	1	1	1	1	0	1	0	1
4	3	3	4	3	3	1	Youtube	1	1	1	1	1	1	1	0	1
3	3	3	4	4	4	0		1	1	0	0	0	0	1	0	1
5	3	3	3	3	3	1	Youtube, Chatgpt, Tiktok	3	1	0	1	1	1	0	0	1
3	4	3	3	3	3	1	Youtube	3	1	1	1	0	1	1	0	1
3	4	4	3	3	3	0		3	1	1	1	1	1	1	0	1
3	3	4	4	3	2	1	Youtube, Tiktok	1	1	1	1	0	1	0	0	1
5	5	3	2	4	4	0		1	1	1	1	1	1	1	0	1
4	4	1	5	4	4	1	A.I	3	0	0	0	1	0	1	0	0
2	1	3	2	3	2	0		3	0	1	0	1	0	0	1 - Hints when stuck in specific level	0
4	4	2	3	2	2	0	Youtube, A.I	1	0	0	1	0	1	1	0	1
3	2	3	4	3	4	1	Youtube, A.I	1	1	1	1	1	1	1	0	1

B7_PRACTICE _MORE	B7_VISUAL_H ELP	B7_LESS_INTI MIDATING	B7_PEER_PR OGRESS	B7_PREPARE _INTERVIEW	B7_DISTRACT ING	B7_PREFER_I RADITIONAL
	1					
0	1	0	1	1	0	0
1	1	0	0	0	1	0
1	0	0	0	1	1	1
1	0	0	0	0	0	0
0	1	1	1	1	1	0
0	1	0	0	0	1	0
1	0	1	0	1	0	0
1	1	1	1	1	0	0
1	1	1	0	1	0	0
1	1	1	0	0	0	0
1	0	0	0	0	0	0
1	0	1	1	1	0	0
1	1	0	0	0	0	0
1	0	0	0	1	0	0
0	1	0	0	1	0	0
0	1	1	0	0	0	0
1	1	1	0	1	0	0
1	1	1	0	1	0	0
1	1	1	1	1	1	1

APPENDIX B

REPORT LAYOUT/FORMS

<Sample Report produced by the proposed system>

APPENDIX C

WRITTEN COMMUNICATION



College of Information and Communications Technology
Burgos Street, Bacolod City,
Negros Occidental, Philippines 6100
Tel. (034) 434 4561

May 05, 2026

Mae B. Lodana, PhD TM
CICT Dean
STI West Negros University
Burgos Street, Bacolod City

THRU: 
Ariel P. Sumagaysay, MaEd, MSCS
Bachelor of Science in Information Technology
Program Head

Orville V. Balangue, MSCS
Bachelor of Science in Computer Science
Program Head

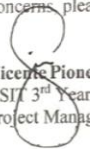
Dear Dr. Lodana

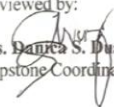
The undersigned are currently working on a capstone project entitled "BALANGKAS: A Gamified E-Learning and Assessment System for Data Structures and Algorithms for STIWNNU 2nd year CICT Students" as part of our academic requirements for the Bachelor of Science in Information Technology. As part of our research process, we intend to conduct a survey among 2nd Year CICT students who have successfully completed the Data Structures and Algorithms (DSA) course during their first semester.

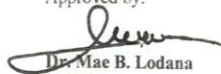
In this regard, we respectfully request your approval to administer this survey to the students. The purpose of this activity is to gather valuable data and insights regarding their past experiences with DSA, which will contribute significantly to the development and evaluation of our gamified system. Rest assured that any information provided by the respondents will be treated with strict confidentiality and used solely for academic purposes.

The survey will be conducted using printed questionnaires and will take approximately 10 to 15 minutes for the students to complete. We will ensure that the distribution and collection of these questionnaires will be properly coordinated so as not to disrupt any of their ongoing scheduled classes.

We would be highly grateful for your favorable response to this request. For any additional questions or concerns, please contact us at 09513622874 or vincepioneta@gmail.com. Thank you.


Vicente Pioneta Jr.
BSIT 3rd Year
Project Manager

Reviewed by:

Ms. Danica S. Duazo
Capstone Coordinator

Approved by:

Dr. Mae B. Lodana
CICT Dean

APPENDIX D

INTERVIEW GUIDE

Capstone Title: BALANGKAS : A GAMIFIED E-LEARNING AND ASSESSMENT SYSTEM FOR DATA STRUCTURES AND ALGORITHMS

Thank you for participating in this survey. This study aims to identify the specific academic and technical roadblocks students face when learning Data Structures and Algorithms (DSA). Your honest feedback will help in developing more effective learning strategies and support systems for IT professionals and students alike.

All responses are anonymous and will be used strictly for research purposes. Please rate the following statements based on your personal experience.

Name (Optional) _____

Year Level: ☐ 1st Year ☐ 2nd Year ☐ 3rd Year ☐ 4th Year

Status: ☐ Regular Student ☐ Irregular Student

Program: ☐ BSIT ☐ BSCS

A. SELF-ASSESSMENT OF PROFICIENCY

1. What is your level of awareness with the concepts of Data Structure and Algorithm?

1.	2.	3.	4.	
	Novice / No perience	Basic Awareness	Working Knowledge	Proficient Ex-
	<i>prior knowledge or exposure to Data Structures and Algorithms (DSA).</i>	<i>Basic awareness of DSA concepts, but limited understanding or experience.</i>	<i>Understands common DSA No concepts and can apply them with some guidance.</i>	<i>Strong understanding of DSA concepts and can confidently apply them to solve problems independently.</i>
	☐	☐	☐	☐

2. How long have you been studying Data Structures and Algorithms (DSA)?

- ☐ Less than a year
- ☐ 1 year
- ☐ More than 1 year
- ☐ Haven't studied at all

3. How difficult is it for you to understand DSA?

- ☐ 1. Very Easy ☐ 2. Easy ☐ 3. Moderate ☐ 4. Difficult ☐ 5. Very Difficult

4. On average, how much time do you spend mastering a specific concept of DSA until you can implement it independently?

- ☐ Less than 1 hour ☐ 1 - 3 hours ☐ A Full Day ☐ Multiple days / Ongoing study

B. PERCEIVED DIFFICULTY AND LEARNING OBSTACLES

1. Provide the difficulty level of each DSA concepts using the scales:

1. Very Easy 2. Easy 3. Moderate 4. Difficult 5. Very Difficult

DSA Concept	Scale	DSA Concept	Scale
1. Arrays		9. Graphs	
2. Linked List		10. Iteration	
3. Stacks		11. Recursion	
4. Queues		12. Algorithms	
5. Trees			
6. Heaps			
7. Sets			
8. Maps			

2. What primary roadblock/s do you face when working and understanding Data Structures and Algorithms? Select All that applies.

- ☐ Understanding Concepts
- ☐ Applying Concepts
- ☐ Visualizing Data Structures
- ☐ Remembering Processes
- ☐ Implementing the Logic
- ☐ Others: _____

3. Please read each statement carefully and select the response that best reflects your personal experience. There are no right or wrong answers; your honest feedback is the most valuable asset to this study.

Scale: [1] *Never* | [2] *Rarely* | [3] *Sometimes* | [4] *Often* | [5] *Always*

	1	2	3	4	5
A. Study Habits & Engagement					

1. I dedicate time to review or practice DSA concepts outside of my required class hours.					
2. I use external resources (e.g., YouTube, AI tools, or textbooks) to supplement my understanding of DSA.					
3. I perform "dry-runs" or manual traces of data structures on paper to understand their behavior.					
B. Technical & Conceptual Roadblocks					

4. I find it difficult to mentally visualize how data moves within a structure (e.g., pointers, recursion, or tree branching).					
5. I understand the theoretical logic of a structure but struggle to implement it in actual code.					
6. I struggle to remember the specific steps or patterns of an algorithm shortly after learning them.					
7. I find it challenging to calculate and solve problems in DSA.					
8. I experience "total blockage" where I cannot proceed with a problem even with the help of hints or notes.					
C. Problem-Solving & Application					
9. I am confident in my ability to choose the most efficient data structure for a new, unseen problem.					
10. I find myself relying on external solutions (e.g., Google or AI) rather than solving the logic independently.					
11. I struggle to identify which specific algorithm (e.g., sorting vs. searching) is appropriate for a given task.					
12. I find the mathematical foundations of DSA more challenging than the actual coding implementation.					

4. Do you have other platforms used when learning DSA?

☐ No ☐ Yes, Specify _____

5. Would gamification (points, badges, levels) motivate you to learn DSA?

☐ Yes ☐ No ☐ Maybe

6. Which gamification features would you prefer? *Select all that applies*

- ☐ Points System
- ☐ Badges/achievements
- ☐ Leaderboards
- ☐ Mini games
- ☐ Progress Tracking
- ☐ Reward System
- ☐ Others: _____

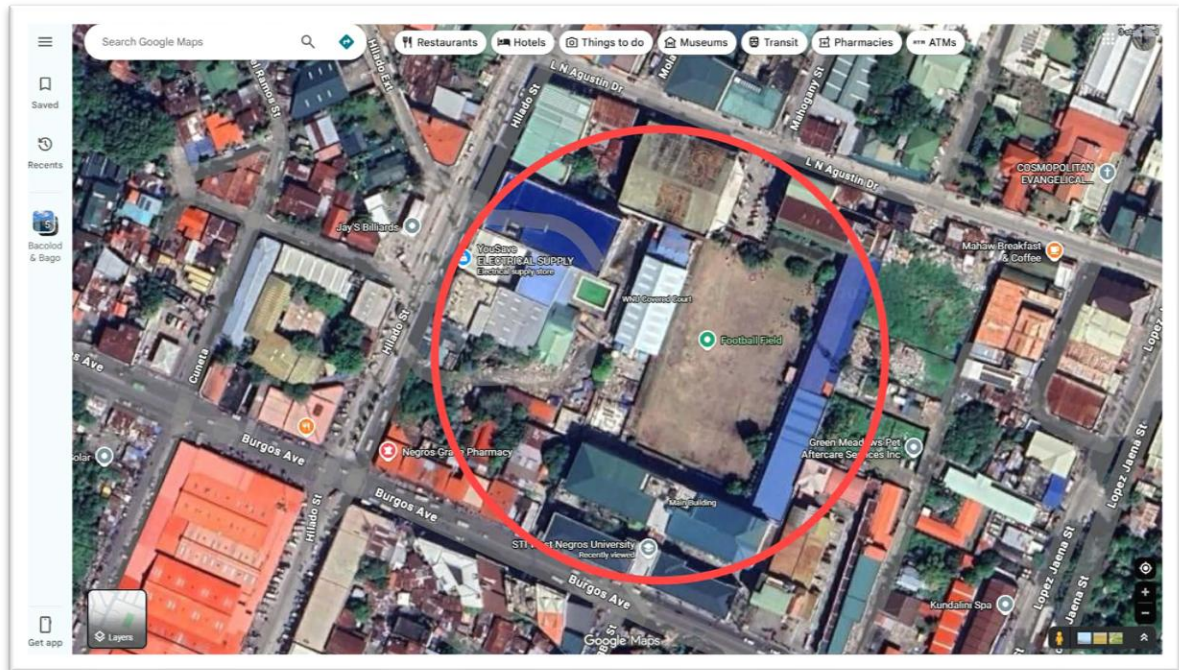
7. If given a chance, what will be your perception of a gamified DSA learning platform? *Check all the applies.*

- ☐ I believe game elements (points, badges, leaderboards) would make me want to practice more consistently.
- ☐ I think the interactive nature would help me "see" and understand complex structures (like Trees and Graphs) better than a textbook.
- ☐ I feel that a game-like environment would make the difficult concepts of DSA feel less intimidating and more approachable.

- ☐ I believe seeing my progress compared to my peers would push me to master topics faster.
- ☐ I think a platform that lets me "level up" through coding challenges would better prepare me for technical interviews.
- ☐ I am concerned that the game elements might distract me from actually learning the deep logic and math required.
- ☐ I believe my learning style is better suited for traditional methods (lectures and labs) rather than games.

APPENDIX E

BENEFFACTOR'S LOCATION MAP



APPENDIX F

PICTURES (DATA GATHERING)

